

Total intended antibiotic delivery related to drug concentration affecting the flow rate of elastomeric devices used in outpatient parenteral antimicrobial therapy (OPAT)

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Abstract

Background: Many studies have investigated drug stability in extended home infusion pumps, but little is known of other factors that may affect the total daily dose the patient receives. It has been noted that up to 40% of these infusions are incomplete after the designated run out time.

Aim: This study investigated whether concentration can affect the flow rate of home infusion pumps, and thus the total dose the patient receives in outpatient parenteral antimicrobial therapy (OPAT).

Method: The flow rates of 10 different drug and dose formulations (4–18 g in 0.9% sodium chloride) in Baxter LV10 (Old Toongabie, Australia) infusion pumps were investigated. The pumps were run out at 31°C and weighed at intervals, with flow rate and total volume infused calculated over the 24-h period.

Results: Piperacillin/tazobactam 18 g was the only formulation not to meet the minimum standard of 90% of the volume being infused over the 24-h period. This indicates a failure of intended therapy. Across all infusions, the expected dose delivered (DD) can be calculated as $\%DD = -1.9x + 122.85$, where x is the total drug and excipient weight within the infusor.

Conclusion: Dose is inversely related to flow rate in the LV10 Baxter infusion pump. Under nominal conditions, any amount greater than 12.02 g in the 240-mL infusion pump will result in an incomplete run out over the 24-h period. Clinicians should be aware that prescribing daily doses higher than 17.28 g may result in <90% of the intended dose being received by the patient. Further research should investigate whether this decrease in flow rate results in poor clinical outcomes.

Keywords: hospital in the home, outpatient parenteral antimicrobial therapy, antibiotic administration, adverse outcomes, infusion pumps, viscosity.

INTRODUCTION

Many studies have investigated drug stability in extended infusions in the outpatient parenteral antimicrobial therapy (OPAT) setting to ensure patients receive their intended dose of therapy.^{1–5} However, few studies have investigated the performance of the devices used or specifically any factors that may affect the total dose patients get in this setting.^{6,7} These other factors may result in larger dose deficits for the patient than drug degradation, and must be understood to ensure best possible patient care.

Elastomeric devices are non-electric mechanical home infusion pumps (MHIP). They are commonly used in

OPAT to administer continuous antibiotic infusions over 24-h periods. Hospital patients with infections requiring extended durations of high-dose intravenous (IV) antibiotics, which are otherwise medically cleared, can be discharged to OPAT services to avoid the higher costs of extended hospital admissions and to relieve pressure on bed spaces.^{8,9} The LV10 Baxter elastomeric pump consists of a 240-mL elastic reservoir within a protective plastic case connected to an IV line with a distal end Luer lock that acts as the flow rate restrictor, designed to release solution at a rate of 10 mL/h. These pumps rely on the pressure of the elastic reservoir to drive the infusion as it deflates. The flow rate restrictor equilibrates to 31.1°C when attached to the periphery of a patient and, at this temperature, is calibrated to infuse the solution at a rate of 10 mL/h when using 5% glucose as the diluent.¹⁰ These devices are stated to run 10% faster when using 0.9% sodium chloride.¹⁰

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The Baxter Clinical Guideline states '*The viscosity of the solution may be affected by the temperature of the solution (drug &/or diluent), and the concentration of the solution thereby impacting the flow rate*'.¹⁰ This statement is not explained further, and no detail is provided on concentration restrictions or limits that allow the device to operate as instructed. Concentration limitations should be described to ensure that the intended dose is provided and expected therapeutic outcomes can be met.

Several studies have noted that the coefficient of viscosity may affect infusion flow rate.^{11–13} However, none of these studies expanded on the relevance to the total dose delivered, nor do the studies detail any limitations that concentration may have on the effective use of the elastomeric devices. Therefore, prescribers may need to be cautious when recommending higher concentrations.

A recent study demonstrated the importance of flow rate with these devices, showing that 40% were not running out to completion within the designated time frame.⁷ Of these devices, 35% had volumes remaining that equated to 17% of the daily dose not being administered.⁷ This warrants further investigation to ensure a better understanding of the optimal functioning of the devices, and therefore leading to better patient care.

The aims of the present study were to investigate the effects of solution concentration on flow rates when using MHIP for commonly used drugs and dosages in OPAT and to determine potential limitations of the dose or drug choice so that the appropriate flow rate is still delivered.

METHODS

Ethics approval was not required for this study because it did not involve humans or animals, or their tissues. The study was on the delivery system used for antibiotics in the OPAT setting.

The LV10 Baxter elastomeric device was chosen for investigation because it is widely used in Australian hospitals. The drugs chosen represent commonly used drugs within the OPAT setting and the doses represent typical total 24-h dosages for patients referred to this service. The antibiotics and strengths investigated were as follows: benzylpenicillin 4.8, 7.2 and 10.8 g; cefoxitin 6 g; ceftazidime 6 g; flucloxacillin 4, 8 and 12 g; and piperacillin/tazobactam 13.5 and 18 g.

Unused LV10 Baxter elastomeric pumps, originally for OPAT patients, were sourced from the hospital pharmacy department. All pumps were used within 2 weeks of the antibiotic expiry and within 2 weeks of the MHIP expiry. Each antibiotic at each dose was investigated in triplicate. All pumps used 0.9% sodium chloride as the

diluent because this is the most common diluent used in the OPAT IV setting.

Change in weight (g) was used as a surrogate marker for volume infused (mL), as seen in previous studies, because the difference in the ratio of mass to volume has been noted to be less than 0.5%.¹⁴

It is estimated that half of all medicines are prepared as salts, and some then need to be made up in buffers to be stable in solution.¹⁵ Therefore, the weight of the active pharmaceutical ingredient within the device could not be used in isolation for comparison of flow rate and volume delivered. The calculations for the weights (drugs and salts and excipients) within each pump set (each antibiotic at each concentration) are given in Table 1.

Pumps were weighed using a Digitech QM-7264 (Jaycar Electronics, Rydalmere, Australia) digital bench scale ((d) = 0.01 g; Class 3). Pumps were weighed prior to flow commencement and then every 4 h until 20 h. After the 20-h point, pumps were weighed every hour until the 24-h point. Hourly measurements were taken in the last 4 h of infusions because the pumps are known to increase flow rate over the last few hours, which may cause them to run out early.¹⁶ The increased flow rate at the end of the infusion is referred to herein as the 'dose dumping phase'.

The pumps were set to run in an incubator (Thermoelectric BCR-25, Changzhou wujin Best Electronic Cables Co., LTD., Changzhou City, China) set at 31°C. Incubator temperature was monitored separately using a Thermocron TCX temperature logger (OnSolution Pty Ltd., Castle Hill, Australia) to ensure strict temperature control. This temperature replicates as close as possible the desired temperature of 31.1°C, which is required to achieve the nominal flow rate for the pumps of 10 mL/h. However, all pumps tested in the present study were in 0.9% sodium chloride, therefore the nominal flow rate is 11 mL/h, as noted above.

Flow rate may change by 0.5% for every 2.54 cm that the distal end of the infusion line sits above or below the reservoir.¹⁰ Therefore, the devices were placed in the incubator ensuring that the distal end Luer lock was at the same level as the reservoir.

The run out period was started by disconnecting the Luer lock caps and allowing the infusion solution to flow into a catchment container within the incubator. These caps were temporarily replaced when the pumps were weighed. The timer was stopped while the caps were on during weighing, and the infusion timer was restarted once the caps had been removed after the pumps had been weighed.

Flow rates were calculated from the change in weight of the pumps at each time point. Residual remaining

Table 1 Common drug doses and total weights within solution

Drug	Daily dose (g)	Drug available as	Drug weight (g/24 h)	Extra excipient ^a weight (g/24 h)	Total weight within solution (g)
Flucloxacillin	4	Flucloxacillin sodium	4.2	0.2	4.4
Benzylpenicillin	4.8	Benzylpenicillin sodium	5.1	0.5	5.6
Cefoxitin	6	Cefoxitin sodium	6.3	0	6.3
Ceftazidime	6	Ceftazidime pentahydrate	7.0	0.7	7.7
Benzylpenicillin	7.2	Benzylpenicillin sodium	7.7	0.8	8.5
Flucloxacillin	8	Flucloxacillin sodium	8.4	0.4	8.8
Benzylpenicillin	10.8	Benzylpenicillin sodium	11.5	1.2	12.7
Flucloxacillin	12	Flucloxacillin sodium	12.6	0.6	13.2
Piperacillin/tazobactam	13.5	Piperacillin sodium/tazobactam sodium	14.1	0.5	14.6
Piperacillin/tazobactam	18	Piperacillin sodium/tazobactam sodium	18.8	0.7	19.5

^aThe excipients are as follows: citrate buffer for flucloxacillin and benzylpenicillin, sodium carbonate for ceftazidime and EDTA and citric acid monohydrate for piperacillin/tazobactam.

volume (mL) was also calculated for each set of pumps using starting weights (g), 24-h weights (g) and the known device volume of 240 mL, as follows:

$$\begin{aligned} \text{Residual volume} = & \text{final weight of each pump} \\ & - (\text{mean weight of all full pumps} \\ & - 240) \end{aligned}$$

Flow rates were calculated, and these rates were plotted against total drug and excipient weight within the infusion.

To analyse the results, the following calculations were performed.

1. The percentage mean volume delivered was calculated for each set of pumps. If the mean volume delivered over the 24-h period was <90% of the intended volume, then this would be considered a failure of intended treatment.
2. The volume delivered during the first 20 h (prior to the dose dumping phase) was plotted against the total weight of the drug plus excipients.
3. The volume delivered during the dose dumping phase was plotted against the total weight of the drug plus excipients.
4. By combining the equations from Steps 2 and 3 above, a maximum amount of drug was calculated that still allows the device to function within the expected standard of delivering a minimum of 90% of the intended volume or dose.

RESULTS

Thirty pumps were tested using five different drugs across nine different doses for a total of 10 different

drug-dose combinations. The mean ($\pm$ SD) weight for full pumps was 303.71 ± 0.93 g. The mean incremental flow rates mapped across a 24-h period are shown in Figure 1. The highest mean ($\pm$ SD) incremental flow rate was for the cefoxitin 6 g pumps (14.85 ± 1.82 mL/h) and lowest, which was not affected by the ending of an infusion mid-way through the increment, was for the piperacillin/tazobactam 18 g pumps (8.17 ± 0.19 mL/h).

The weight change of the pumps over the 24-h period is shown in Figure 2. The residual volume in each pump was calculated, and the mean volumes for each pump are given in Table 2.

All sets of pumps contained residual volume at the end of the 24-h period. In Table 2, this is expressed as a percentage of infusion not delivered. The three drugs with the highest percentage of residual volume were flucloxacillin 12 g and piperacillin/tazobactam 13.5 and 18 g (2.41%, 5.08% and 14.15%, respectively).

Flucloxacillin 4 g was the lowest dose tested and the first to finish its run out by the 22-h weighing point. All pumps finished releasing solution by the 24-h point except for the flucloxacillin 12 g and the piperacillin/tazobactam 13.5 and 18 g pumps.

Mean volumes from the first 20 h of infusion were plotted against total drug and excipients within the pumps (Figure 3). The slope of the graph (y) shows an inverse relationship between the drug plus excipients and the volume delivered to give the following equation:

$$y = -3.1596x + 229.94 (R^2 = 0.991) \quad (1)$$

In Eqn 1, y is the volume delivered (mL) and x is the weight of the drug and excipients (g). (Note, y and x represent the same parameters in subsequent equations.)

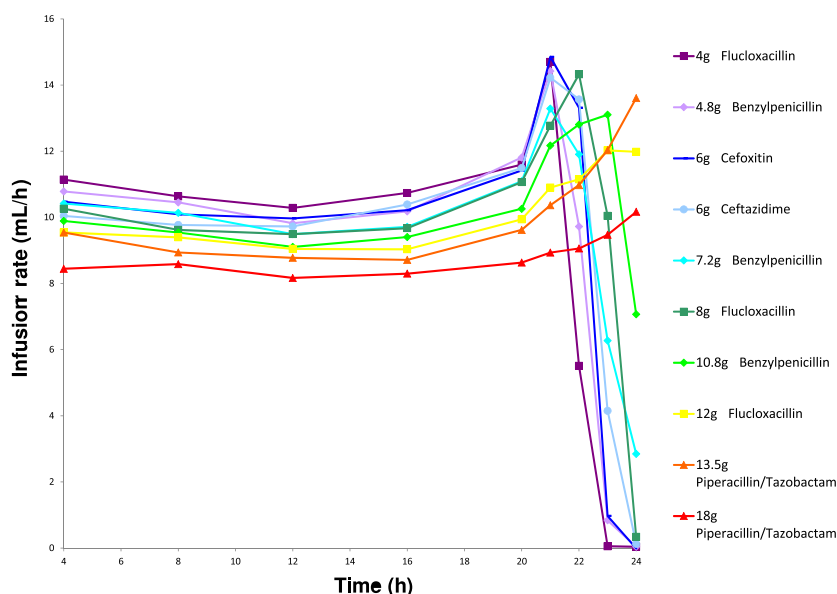


Figure 1 Mean flow rates of the Baxter LV10 infusion pumps run out over a period of 24 h.

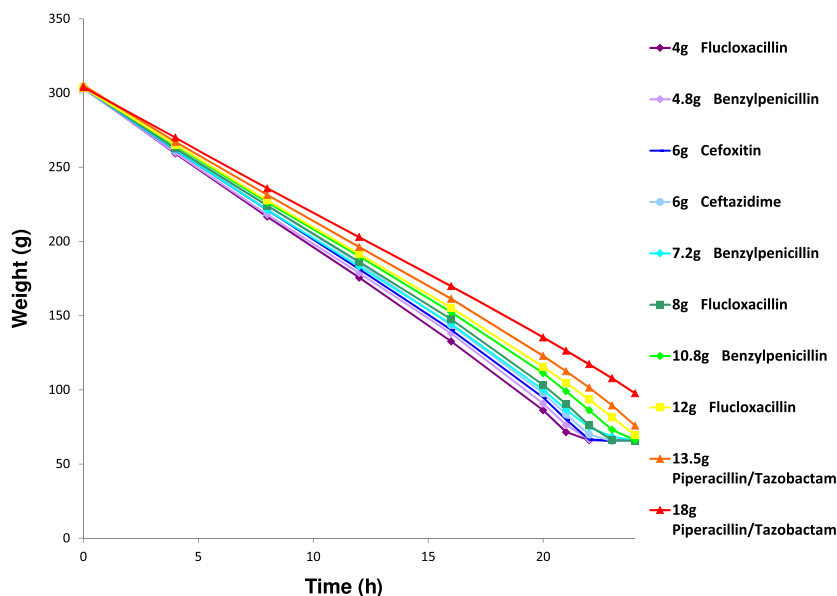


Figure 2 Mean weights of Baxter LV10 infusion pumps run out over a period of 24 h.

Equation 1 was used to determine the weight of drug plus excipients required to achieve a nominal device flow rate of 10 mL/h for the first 20 h and thus deliver 200 mL. The threshold under which flow was not impeded by concentration was calculated to be 9.47 g. It is known from the manufacturer's specifications that the last 4 h of MHIP flow demonstrates an increase in rate during the dose dumping effect. Therefore, Eqn 1

suggests that all devices with less than 9.47 g total drug and excipient weight will result in complete delivery of the solution under nominal device conditions in 0.9% sodium chloride.

Calculation of the dose delivered during the final dose dumping phase requires an additional formula. This final phase was calculated by plotting the mean volume delivered in the last 4 h against total drug plus

Table 2 Mean residual volumes left in elastomeric Baxter LV10 pumps after a 24-h run out period

Drug in 0.9% sodium chloride	Dose (g)	Residual volume (mL)	% Infusion not delivered
Flucloxacillin	4.0	2.35	0.98
Benzylpenicillin	4.8	2.15	0.90
Cefoxitin	6.0	2.05	0.85
Ceftazidime	6.0	2.02	0.84
Benzylpenicillin	7.2	2.08	0.87
Flucloxacillin	8.0	2.17	0.90
Benzylpenicillin	10.8	2.52	1.05
Flucloxacillin	12.0	5.90	2.46
Piperacillin/tazobactam	13.5	12.31	5.13
Piperacillin/tazobactam	18.0	34.08	14.20

Bolded values indicate incomplete runout of infusion

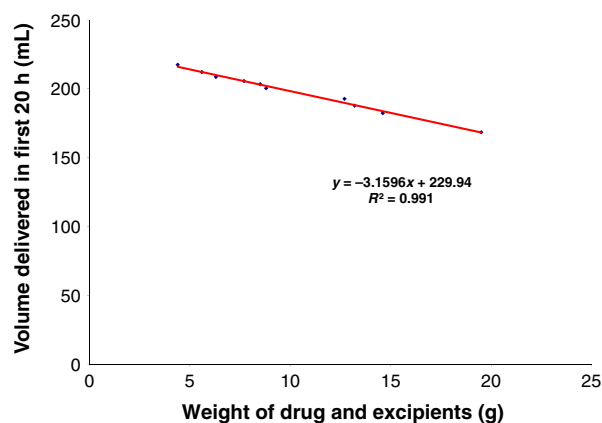
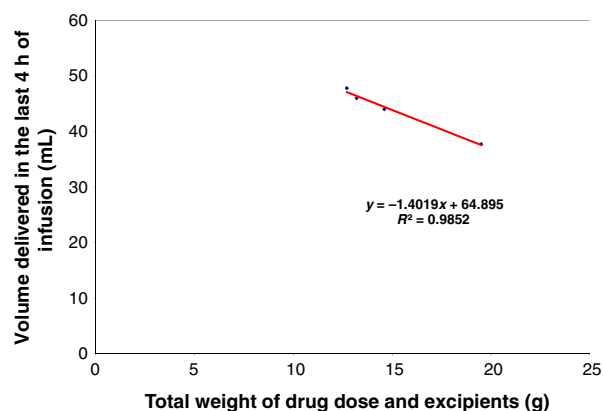
excipient weight if the dose was greater than 9.47 g (Figure 4). From Figure 4, the inverse relationship between the drug plus excipient versus the volume delivered can be expressed as follows:

$$y = -1.4019x + 64.895 (R^2 = 0.9852) \quad (2)$$

Combining Eqn 1 and Eqn 2, the slope (y) can be calculated for the Baxter LV10 infusor over 24 h using sodium chloride 0.9% under nominal conditions as follows:

$$\begin{aligned} y &= -3.1596x + 229.94 - 1.4019x + 64.895 \\ \rightarrow y &= -4.4615x + 294.835 \end{aligned} \quad (3)$$

Using Eqn 3, the maximum drug and excipient weight to achieve a complete delivery of solution

**Figure 3** Volume delivered in the first 20 h according to total drug and excipient weight.**Figure 4** Volume delivered in the last 4 h of infusion for total drug and excipient weights greater than 9.47 g.

(240 mL) within the 24-h period is 12.02 g:

$$\begin{aligned} y &= -4.4615x + 294.835 \text{ (where } y = 240\text{)} \\ \text{so, } 240 &= -4.4615x + 294.835 \\ -54.835 &= -4.4615x \\ x &= 12.02 \end{aligned}$$

In addition, using Eqn 3, the drug and excipient weight to achieve the minimum delivery of 90% of the intended dose (216 mL) within the 24-h period is 17.28 g:

$$\begin{aligned} y &= -4.4615x + 294.835 \text{ (where } y = 216\text{)} \\ \text{so, } 216 &= -4.4615x + 294.835 \\ -78.835 &= -4.4615x \\ x &= 17.28 \end{aligned}$$

Doses over 12.02 g should use $y = -4.4615x + 294.835$ to calculate the volume delivered. Alternatively, a more useful reference is to calculate the percentage dose delivered by the following equation:

$$\begin{aligned} \% \text{Dose delivered} &= (-4.4615x + 294.835) / 240 \times 100 \\ \% \text{Dose delivered} &= (-4.4615x + 294.835) / 2.4 \\ \% \text{Dose delivered} &= -1.9x + 122.85 \end{aligned} \quad (4)$$

DISCUSSION

This study demonstrates that the 24-h flow rate of elastomeric pumps mimics the pattern described by the manufacturer.¹⁶ The pattern showed that the flow rate slowed in the middle phase of the infusion (at ~12 h), then increased substantially over the dumping phase (at ~22 h). Although the pattern of change in flow rate in

this study was similar to the manufacturer's specifications, there was a noticeable difference in the overall rate (Figure 1).

All pumps had residual solution remaining in the device. The MHIPs with doses from 4 to 10.8 g had a mean ($\pm$ SD) percentage residual volume of $0.87 \pm 0.07\%$. Less than 1% of solution is likely to remain within the deflated balloon reservoir at the end of the run out period and is therefore unlikely to be linked to an incomplete infusion. The MHIPs with higher-concentration antibiotics (e.g. flucloxacillin 12 g and the piperacillin/tazobactam 13.5 and 18 g) had residual volumes of 2.41, 5.08 and 14.15%, respectively. These residual volumes are more likely to imply an incomplete infusion. Incomplete infusions have been confirmed as a problem in the OPAT setting in previous studies.⁶ The present study demonstrates that the higher doses may be a contributing factor.

Piperacillin/tazobactam 18 g was the only MHIP to breach the 90% threshold. Less than 90% of the volume delivered indicates a failure of intended dosing, which has the potential to result in patient treatment failure (e.g. failed antimicrobial eradication, extended courses of antibiotics or readmission to hospital). Further investigation into the clinical relevance of the decrease in volume delivered is recommended.

Our data show flow rates of drugs using 0.9% sodium chloride as the diluent, not 5% glucose. The manufacturer specifications state that 5% glucose as the diluent causes a 10% slower flow rate. Using glucose as a diluent may mean other doses of drug infusions may breach the 90% threshold. However, flow rates using glucose need to be investigated appropriately to confirm this assumption.

An inverse relationship was found between the weight of the drug and excipients within the LV10 devices and the flow rate for the pre-dose dumping phase and the dose dumping phase. This study demonstrates that dose does affect flow rate, and so prescribers should be aware that higher doses of drugs may result in less than 90% of the volume of the infusions being delivered.

Under nominal device conditions, complete dose delivery will occur for all doses where the dose and excipients add up to ≤ 12.02 g. All doses above this weight will result in less than 100% of the dose being received to the patient. For these regimens, the percentage of the intended dose that will be delivered can be calculated as follows: % dose delivered = $-1.9x + 122.85$.

The maximum dose plus excipient weight that will achieve the minimum dose delivery of 90% of the intended volume is 17.28 g. Doses above this will result in a failure of intended treatment.

To the best of the authors' knowledge, the present study is the first to find a relationship between the total dose within these devices and the resulting flow rate and dose delivery. Further studies should investigate whether other factors place limitations on the use of these devices in the OPAT setting.

Weighing the devices more frequently may have provided more detail on the transition from pre-dose dumping to dose dumping phases, and thus a more accurate final equation. Other study limitations include using devices and drugs within 2 weeks of expiry. This may have the potential to affect the final results. The authors are also aware that these results are reflective of volume of infusion, and thus the dose being delivered only, not actual clinical outcomes.

The clinical relevance of the resulting decrease in volume delivered for higher-dose infusions should be investigated thoroughly to further guide clinicians. However, in light of the lack of any further clinical evidence, clinicians should avoid exceeding the limitations described in this study where possible.

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Conflicts of interest statement

Stephen Perks has previously received grant moneys for other research from Baxter Healthcare Corporation, which is a manufacturer of mechanical infusion devices. The other authors have no conflicts of interest to declare.

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